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Effects of contralateral versus ipsilateral electroacupuncture for analgesia and rehabilitation after unilateral total knee arthroplasty: a randomized controlled trial

Hai Huang[1†], Kangmin Tang[2†], Xiuling Song[2], Ling Zhao[2], Yongying Liang[3], Hui Xu[3], Lianbo Xiao[3] and Yuelai Chen[2,4]

Abstract

Purpose: Total knee arthroplasty (TKA) is a treatment for advanced knee osteoarthritis. Since postoperative pain affects rehabilitation, this study aimed to determine whether electroacupuncture (EA) contralateral to the surgical site is more effective than ipsilateral EA or sham EA in terms of relieving postoperative pain and promoting post-TKA rehabilitation.

Methods: In this parallel, single-blind randomized controlled trial, 114 patients undergoing unilateral TKA were assigned to the contralateral EA (EA on the contralateral side + sham EA on the ipsilateral), ipsilateral EA (EA on the ipsilateral + sham EA on the contralateral side), or sham EA (sham EA on both sides) groups (n = 38 each). Treatment was performed once daily on postoperative days 1–3. The visual analog scale (VAS) scores, additional opioid doses via patient-controlled analgesia (PCA) pump, Hospital for Special Surgery (HSS) knee scores, active/passive range of motion (AROM/PROM), swelling around the knee joint, and Hamilton anxiety scale (HAMA) scores were used for postoperative evaluation.

Results: At 3 days postoperatively, the VAS scores, HSS scores, AROM/PROM, swelling around the knee, and HAMA scores in the contralateral EA and ipsilateral EA groups were significantly improved compared with baseline. In addition, VAS scores, HSS scores, PROM and swelling around the knee were significantly better in the contralateral and ipsilateral EA groups than in the sham EA group, but similar in the two true EA groups. Furthermore, PCA additional dose release was significantly higher in the sham EA group than in the two true EA groups (which did not significantly differ). At 10 days postoperatively, the HSS scores, AROM/PROM, and HAMA scores were better in the contralateral and ipsilateral EA groups than in the sham EA group, but similar in the two true EA groups.

Conclusion: Contralateral EA is more effective than sham EA for treating postoperative pain following TKA, but has an analgesic effect similar to that of ipsilateral EA.

Trial registration number: ChiCTR1800020297 (Chinese Clinical Trial Registry)

Keywords

contralateral needling, electroacupuncture, knee osteoarthritis, ipsilateral needling, pain management

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[1]Shenzhen Hospital, Shanghai University of Traditional Chinese Medicine
[2]Shanghai University of Traditional Chinese Medicine, Shanghai, China
[3]Guanghua Hospital, Shanghai University of Traditional Chinese Medicine, Shanghai, China
[4]Longhua Hospital, Shanghai University of Traditional Chinese Medicine, Shanghai, China

†Hai Huang and Kangmin Tang contributed equally to this work and should be considered co-first authors.

Corresponding authors:
Lianbo Xiao, Guanghua Hospital, Shanghai University of Traditional Chinese Medicine, Shanghai 200052, China.
Email: xiao_lianbo@163.com

Yuelai Chen, Longhua Hospital, Shanghai University of Traditional Chinese Medicine, Shanghai 200032, China.
Email: chen_yuelai@163.com

Introduction

Acupuncture is widely used in the treatment of knee osteoarthritis (KOA). Acupuncture can effectively improve the pain and joint issues associated with early KOA and also play an important role in reducing postoperative pain after total knee arthroplasty (TKA) in patients with late KOA. It has been reported that approximately 7 million living people in the United States have had a total hip arthroplasty or TKA.[1] With the aging global population, the number of TKA surgeries performed in most countries, including China, has increased annually, as has the economic burden of medical care. However, it has been reported that 15%–20% of patients undergoing TKA are unsatisfied with their postoperative outcome,[2,3] while joint pain after TKA is known to affect the rehabilitation process and patient satisfaction.[4] About 30% of patients experience moderate postoperative knee pain, and 60% experience severe postoperative knee pain.[5] Acute pain that is not promptly and effectively controlled greatly increases the risk of postoperative complications and persistent pain.[6] The widespread use of prescription opioids has led to an increasing number of people receiving these agents.[7,8] Presently, the most common postoperative analgesic approach is multimodal analgesia,[9] which can effectively improve postoperative outcomes and reduce opioid consumption.[10,11] However, the use of any one or more medications is associated with side effects (such as psychological dependence and gastrointestinal-related events),[12] and it has become essential to develop and implement multidisciplinary methods to improve TKA outcomes while reducing unnecessary expenditure.[13] In recent years, nonopioid medications and alternative therapies have become prevalent. Electroacupuncture (EA) can reduce pain after TKA without side effects and thus influence the recovery period.[14,15]

A meta-analysis previously demonstrated that postoperative pain after TKA was reduced with the use of EA.[16] Contralateral acupuncture, also known as opposing needling or healthy side acupuncture, originally described in the *Huangdi Neijing*, was one of the nine needling methods described in ancient China.[17,18] In this acupuncture method, to avoid infection near the incision after TKA, traditional acupuncture point locations on the healthy side are selected for treatment. Historically, contralateral acupuncture has been widely used in the clinical treatment of pain and joint dysfunction in China, and in recent years, increasing research has been focusing on this method, as it can relieve chronic shoulder pain and improve shoulder mobility.[19–21] Studies have shown distinct antinociceptive effects both from contralateral and ipsilateral needling, and the former may exert an analgesic effect through spinal and supraspinal mechanisms.[19,22] Accordingly, we conducted a randomized controlled trial to compare the short-term effects of contralateral EA (versus ipsilateral EA and sham EA) on pain after TKA, and evaluated whether contralateral needling can accelerate rehabilitation after TKA.

Methods

Ethical approval

The protocol for this clinical trial was in accordance with the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) guidelines and has been previously published.[23] This study was assessed and approved by the Institutional Review Board of Guanghua Hospital, Shanghai University of Traditional Chinese Medicine (2018-K-18) and was prospectively registered in the Chinese Clinical Trials Registry Center (ChiCTR1800020297) on 22 December 2018 prior to recruitment of the first participant. All patients provided written informed consent before randomization.

Study design and enrollment

This was a randomized, controlled, three-arm parallel trial comparing the efficacy of two interventions (contralateral EA vs ipsilateral EA) relative to a control intervention (sham EA). All participants were inpatients admitted to the Joint Surgery Department of Guanghua Hospital, Shanghai University of Traditional Chinese Medicine, between February and November 2019.

Randomization and masking

Patients with KOA were enrolled, and unilateral TKA was performed as planned after admission. The day before the surgery, baseline assessment of eligible patients was carried out and written informed consent was obtained. The patients were then randomly assigned (in a ratio of 1:1:1) to an contralateral EA group, ipsilateral EA group, or sham EA group. Random numbers were generated using the Statistical Package for the Social Sciences (SPSS; IBM Corp, Armonk, NY, USA) before the trial and sealed in opaque envelopes. Only the acupuncturist had the right to open the envelope to obtain the group code for the corresponding intervention. All other researchers, such as assessors and statistical analysts, remained blind to group allocation.

Participants

To be included, patients (of either sex) needed to be: aged 40–75 years; diagnosed with KOA according to the clinical classification criteria recommended by the American College of Rheumatology; and undergoing unilateral TKA under general anesthesia during hospitalization without surgical and anesthetic contraindications.

We excluded patients who: could not participate due to skin lesions at the planned sites of acupuncture needling; had lower extremity sensory disturbance or abnormality; had severe arrhythmias, heart failure, chronic obstructive pulmonary disease, epilepsy, or mental disorders; or had received acupuncture treatment during the month before enrollment.

Interventions

All patients underwent surgery under general anesthesia. The anesthetic drugs included fentanyl (4 μg/kg), propofol (2 mg/kg), and cisatracurium (0.15 mg/kg), and anesthesiologists paid close attention to each patient's physical condition. Each patient received the same analgesic dose (continuous infusion rate 0.25 g/kg/h) via a patient-controlled analgesia (PCA) pump after surgery. Furthermore, when the patient's pain severity was unbearable, and the visual analogue scale (VAS) score was >60,[24] additional analgesics (etoricoxib tablets, 60 mg) were provided according to the patient's needs after discontinuing the PCA pump.

The acupuncturists in the research process had many years of clinical experience and completed unified training for the research program. The results were evaluated by research evaluators, and the research process was monitored by the main researchers. The treatment of the three groups included three courses within the 3 days after surgery.

Based on previous research experience and expert discussion, EA was administered at ST32 (*Futu*) and ST36 (*Zusanli*), featuring a distribution around the knees, as well as SP9 (*Yinlingquan*) and GB34 (*Yanglingquan*) (Supplemental Figure 1).

During each 30-min acupuncture session, the patient was placed in a supine position. The acupuncturist untied the elastic bandage that had been applied during the operation to expose the treatment area. The local skin was sterilized with a cotton ball dipped in 75% alcohol, and then, a sterile rubber pad (10 mm diameter and 5 mm thickness) was pasted onto each traditional acupuncture point location. For EA, four verum needles (0.30 mm in diameter and 40 mm in length; Hwato Brand, Suzhou Medical Appliance Factory, China) were inserted vertically through the sterile rubber pads to a depth of 25–30 mm, until a muscle twitch response was elicited. Subsequently, the two connecting wires of an EA instrument (SDZ-V, Hwato Brand) were connected to the handles of needle pairs GB34 + ST32 and ST36 + SP9, which were stimulated with a continuous wave of 2 Hz frequency, 1 mA intensity, and 0.25 ms pulse width. For sham EA, four blunt needles (0.30 mm in diameter and 25 mm in length; Hwato Brand) were used to touch the skin through the sterile rubber pad without skin penetration, and the needle handles were rotated on both sides three times to simulate real acupuncture. The internal metal wire of the EA instrument was cut, such that no current passed through the connected power line; however, the display of the EA unit was identical to that observed by the patients in the true EA groups.

Contralateral EA. This group received EA on the healthy side of the lower limb and sham EA on the ipsilateral.

Ipsilateral EA. This group received EA on the same side as the operated lower limb and sham EA on the healthy side.

Sham EA. This group received sham EA on both the operated and the healthy sides.

Outcome measurements

Index measurement time points were postoperative days 1–3 days during the treatment period, and postoperative days 5, 7 and 10 during the follow-up period (Supplemental Table 1).

The two prespecified primary outcome measurements were as follows: (1) pain, assessed using the VAS, which was evaluated after EA treatment when patients were walking or moving their joints on six different occasions (postoperative days 1, 2, 3, 5, 7 and 10); and (2) the additional dose released by PCA pump as recorded on postoperative day 2 (i.e. 24 h after the first treatment).

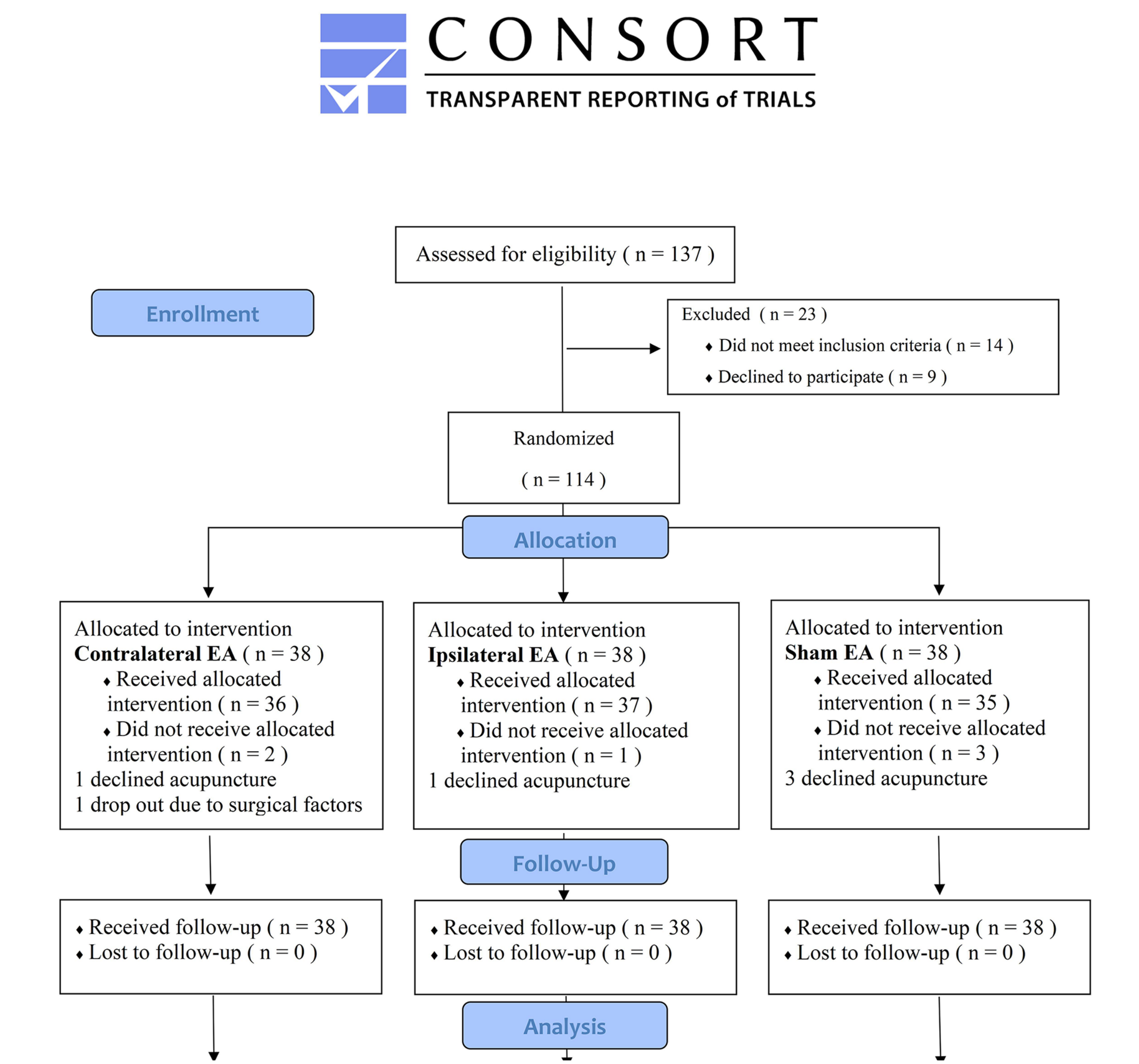
The secondary outcome measures were the Hospital for Special Surgery (HSS) knee and Hamilton anxiety scale (HAMA) scores, which were evaluated on postoperative days 3 and 10. The knee joint's active and passive range of motion (AROM and PROM) and improvement of swelling around the knee joint were evaluated on postoperative days 3, 5, 7, and 10. The latter was based on the knee circumference ratio on postoperative day 1, according to the following formula: (circumference of the knee (COK) on day X—COK on postoperative day 1) / (COK on postoperative day 1). During the first measurement, a mark was made at the suprapatellar pole and used as the starting point over the next few days when measuring the COK around the patella with a tape measure.

Furthermore, any side effects of acupuncture (such as pain, local infection, fatigue or local hematoma)[25] and acupuncture tolerance,[26] including discomfort and acceptance of EA, were also evaluated on postoperative days 1–3, and the blinding success rate was assessed. Cases that dropped out after receiving 1–2 sessions of EA or sham EA were regarded as blinding failures, consistent with a prior trial by Liu et al.[27] Subjects completed their blinding assessments by means of a questionnaire, which was administered within 5 min of the end of the three interventions.[23]

Sample size calculation and statistical analysis

The strategy of sample size calculation was performed based on the mean ± standard deviation (SD) for a pilot sample, in which contralateral side EA, ipsilateral EA, and sham EA yielded VAS scores of 38.6 ± 9.9, 45.4 ± 8.0, and 46.9 ± 7.4, respectively. Assuming SD = 9.9, two-tailed $\alpha = 0.05$, 90% power ($1-\beta$), and $\lambda = 12.66$ (noncentral parameter of the κ-1 noncentral chi-square distribution),[28] it was estimated that a total sample size of 96 cases (n = 32

Figure 1. CONSORT flow diagram of patients.



per group) was needed. Considering an anticipated dropout rate of 15%, target recruitment per group was increased to n=38 (total sample size of 114). After the study, the final mean and SD were transferred back into the sample size calculation formula, and it was concluded that each group had only needed only 33 subjects to achieve 90% power, indicating that the assumptions of the *a priori* sample size calculation had been appropriate.

Data were analyzed by a statistician using the Statistical Package for the Social Sciences (SPSS) version 26.0 (IBM Corp, NY, USA), and analysis was based on the intention-to-treat (ITT) principle.[29] Continuous data are expressed as the mean and SD or median and interquartile ranges as appropriate, and categorical data are expressed as numbers and percentages. For normally distributed data, the baseline analysis was performed using the t-test or Wilcoxon rank-sum test for continuous data and chi-square test or Fisher's exact test for categorical data. Comparisons between the three groups were conducted using one-way analysis of variance (ANOVA) and post hoc tests of least significant difference (LSD).[30] When the distribution was not normal, the Kruskal–Wallis test was used. The preoperative data were used as covariates, and the primary outcome was assessed using a one-way, repeated-measures ANOVA to test the influence of time, group, and time × group interaction on pain. Comparisons of categorical data between groups were made using the chi-square test or the Mann–Whitney U-test as appropriate. Given our seven primary outcomes (VAS score at six different time points plus the additional dose delivered by PCA (assessed once on postoperative day 2), $p < 0.007$ was taken to indicate formal statistical significance (i.e. $p < 0.05$ with Bonferroni adjustment to account for more than one primary outcome). In addition, a Sidák correction was applied to control for the multiple pairwise testing associated with three study groups. Statistical significance for our secondary outcome measures was kept at $p < 0.05$, on the understanding that these were exploratory in nature.

Results

Patient characteristics

Between February and December 2019, we screened 137 participants, 114 of whom were randomly assigned to treatment groups. Of these participants, 108 (95%) completed the study. Six participants dropped out (two from the contralateral EA group, one from the ipsilateral EA group, and three from the sham EA group). The dropout rate and the number of primary outcome assessments completed at each stage are shown in Figure 1. Table 1 shows the general and baseline characteristics of the participants, which were comparable among the three groups.

Primary outcomes

Pain. There were no significant differences observed in the VAS score between the three groups at baseline (Table 2). Figure 2 shows a downward trend in the VAS scores in the contralateral EA and ipsilateral EA groups during the treatment and follow-up periods. After three EA interventions (postoperative day 3), the VAS scores were significantly lower in both the contralateral EA and ipsilateral EA groups compared with the sham EA group ($p < 0.001$); this trend continued on postoperative days 5, 7, and 10. The VAS scores did not significantly differ between the contralateral EA and ipsilateral EA groups at any point during the treatment and follow-up periods.

Additional doses delivered by PCA pump. On postoperative day 2 (i.e. 24 h after the first intervention), the additional dose delivered by the PCA pump was greater in the sham EA group than in the contralateral EA group ($p < 0.001$). While a similar trend was seen in ipsilateral EA vs sham EA groups, it did not meet our adjusted threshold for statistical significance ($p < 0.05$ but >0.007; Table 2).

Secondary outcomes

Knee joint function. After three EA interventions, the HSS scores were higher in the contralateral EA and ipsilateral EA groups than in the sham EA group; however, scores did not significantly differ between the contralateral EA and ipsilateral EA groups ($p < 0.01$). Furthermore, the HSS score improvement was greater in the contralateral EA and ipsilateral EA groups than in the sham EA group on postoperative day 10 (both $p < 0.001$; Table 3).

Range of motion. The range of motion (ROM) assessment included AROM and PROM (Table 3). After three EA interventions, the PROM in the contralateral EA group was greater than that in the sham EA group ($p < 0.05$). PROM was significantly greater during the follow-up period in the ipsilateral EA group than in the sham EA group on postoperative day 5 ($p < 0.05$). Both AROM and PROM in the contralateral EA group were significantly greater than those in the sham EA group on postoperative day 7 ($p < 0.05$). AROM and PROM in both the contralateral EA and ipsilateral EA were significantly greater than those in the sham EA group on postoperative day 10 ($p < 0.05$).

Improvement in swelling around the knee joint. After three EA interventions, the improvement in the swelling around the knee joint was significantly greater in the ipsilateral EA group compared with both the contralateral EA ($p < 0.05$) and sham EA ($p < 0.001$) groups. On postoperative day 5, the swelling had gradually reduced, but the improvement was still greater in the ipsilateral EA group than in the sham EA group ($p < 0.05$).

Anxiety. After three EA interventions, the HAMA score was significantly lower in the ipsilateral EA group than in the sham EA group ($p < 0.01$). On postoperative day 10, the HAMA scores of the contralateral EA and ipsilateral EA groups were both significantly lower than in the sham EA group ($p < 0.01$ and $p < 0.001$, respectively).

Safety and other outcomes

Three cases of side effects of acupuncture were recorded (Table 4). Two of these were in the contralateral EA group; a bent needle occurred in one patient, and one patient became slightly flustered before acupuncture. Meanwhile, one patient in the ipsilateral EA group had a curved needle. Our experienced acupuncturist pacified the patient, removed the acupuncture needle, and gave the patient a cup of warm boiled water, and they felt relieved after a short rest.

Acupuncture tolerance was evaluated and it was found that the discomfort of EA decreased while the acceptance of EA increased day by day, with no significant differences between groups (Table 4).

Finally, the total number of blinded cases was 85, and the blinding success rate was 75% (Table 4).

Discussion

Pain management after TKA is an urgent problem to be solved in patients with KOA,[9,31] as postoperative pain trajectories are independently associated with long-term pain

Table 1. General and preoperative baseline characteristics of participants in the three groups.

Variable		Contralateral EA (n = 38)	Ipsilateral EA (n = 38)	Sham EA (n = 38)	P
Age, years		67.29 ± 4.78	68.82 ± 5.36	67.58 ± 4.95	0.377
Gender					
Female		33 (86.8)	36 (94.7)	31 (81.6)	0.251 (χ^2 = 3.107)
Male		5 (13.2)	2 (5.3)	7 (18.4)	
Height, m		1.58 [1.55, 1.60]	1.58 [1.54, 1.64]	1.56 [1.54, 1.61]	0.470
Weight, kg		61.50 ± 8.94	61.26 ± 10.44	58.84 ± 10.40	0.439
BMI		24.43 ± 3.60	24.12 ± 3.69	23.34 ± 2.92	0.366
Medical history					
Yes		22 (57.9)	25 (65.8)	26 (68.4)	0.609 (χ^2 = 0.990)
No		16 (42.1)	13 (34.2)	12 (31.6)	
Operative time, h		1.38 ± 0.36	1.38 ± 0.36	1.41 ± 0.30	0.934
TKA laterality					
Left		18 (47.4)	21 (55.3)	18 (47.4)	0.729 (χ^2 = 0.632)
Right		20 (52.6)	17 (44.7)	20 (52.6)	
Baseline characteristics					
VAS[a]		37.18 ± 12.05	37.82 ± 11.03	39.71 ± 10.43	0.593
Total dose of PCA pump (mL)		100	100	100	1.000
ROM[a]	AROM	90.00 [85.00, 100.00]	90.00 [80.00, 101.25]	90.00 [85.00, 100.00]	0.973
	PROM	100.00 [93.75, 110.00]	100.00 [90.00, 110.00]	100.00 [90.00, 110.00]	0.741
HSS score[a]		53.75 ± 8.38	55.15 ± 8.61	55.72 ± 7.38	0.557
HAMA score[a]		19.16 ± 3.98	18.00 ± 4.56	18.24 ± 2.83	0.388
COK[b]		40.89 ± 3.02	40.06 ± 2.92	41.11 ± 2.96	0.273

Note: Data are mean ± SD, median [Q1, Q3], or n (%).
Baseline characteristics [a]before surgery or [b]before treatment.
BMI: body mass index; VAS: visual analog analogue score; PCA: patient-controlled analgesia; HSS: Hospital for Special Surgery; HAMA: Hamilton anxiety scale; COK: circumference of knee; EA: electroacupuncture; ROM: range of movement.

outcomes after TKA. Since the intensity of postoperative acute pain correlates with the risk of developing a persistent pain state,[32,33] active and early treatment of postoperative pain should also be studied. Compared with other alternative therapies, acupuncture appears to have greater potential in the management of postoperative pain after TKA.[34,35]

After TKA surgery, the knee joint on the ipsilateral of the patient was fixed by a pressure bandage and was not suitable for the local application of EA around the knee joint because it increases the risk of local infection. In this case, it is arguably better to perform EA on the contralateral side. Therefore, in our trial, we aimed to compare the effects of contralateral EA, ipsilateral EA, and sham EA on pain after TKA and evaluate the hypothesized superiority of contralateral EA in rehabilitation. ITT analysis revealed that early active EA was effective at treating postoperative pain after TKA; however, the contralateral and ipsilateral EA methods did not differ with respect to the observed reductions in postoperative acute pain and opioid use. Accordingly, both EA approaches appear to protect against pain and promote joint rehabilitation after TKA.

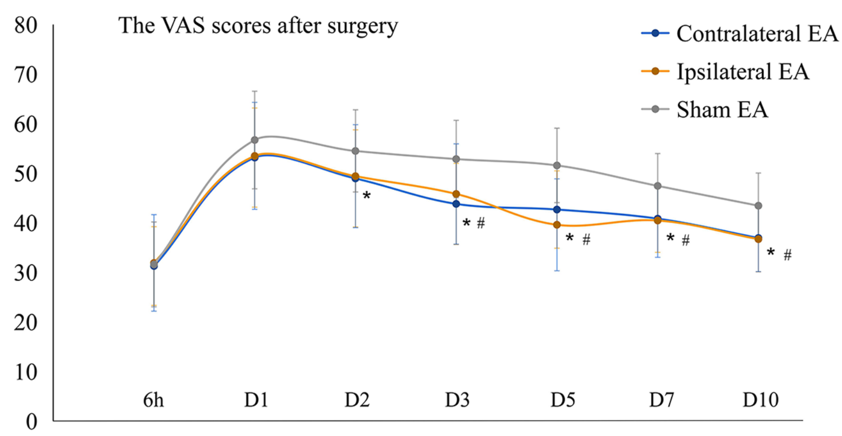
In our trial, the VAS scores did not significantly differ between the two real EA groups during the treatment and follow-up periods but were lower than the sham EA group. Furthermore, this result has also been confirmed in our animal experiments.[36] EA is often used to treat pain in the clinic, but the specific analgesic mechanism is not completely clear. Acupuncture information is introduced from the periphery, reaches the upper center from the spinal cord crossing to the contralateral ventrolateral spinal cord bundle, and induces

Table 2. Characteristics and changes in primary outcomes with multiple comparisons.

Outcome	Contralateral EA	Ipsilateral EA	Sham EA	Pairwise comparisons					
	Group A	Group B	Group C	Group A vs B		Group A vs C		Group B vs C	
				Effect size (95% CI)	P[a]	Effect size (95% CI)	P[a]	Effect size (95% CI)	P[a]
VAS score after surgery									
Day 0[b]	31.29 ± 7.94	31.92 ± 9.74	31.61 ± 8.56	—	—	—	—	—	—
Day 1	53.16 ± 9.99	53.50 ± 10.79	56.71 ± 9.84	−0.28 (−5.97 to 5.41)	0.999	−3.46 (−9.15 to 2.24)	0.445	−3.18 (−8.87 to 2.51)	0.178
Day 2	48.95 ± 9.80	49.42 ± 10.39	54.50 ± 8.29	−0.40 (−5.69 to 4.90)	0.997	−5.43 (−10.73 to −0.14)	0.043	−5.04 (−10.33 to −0.26)	0.067
Day 3	43.82 ± 8.18	45.82 ± 10.11	52.87 ± 7.80	−1.90 (−6.74 to 2.93)	0.715	−8.90 (−13.74 to −4.07)	<0.001	−7.00 (−11.83 to −2.17)	0.002
Day 5	42.68 ± 7.80	39.61 ± 9.28	51.58 ± 7.50	3.17 (−1.37 to 7.71)	0.255	−8.75 (−13.30 to −4.21)	<0.001	−11.92 (−16.47 to −7.38)	<0.001
Day 7	40.84 ± 6.76	40.47 ± 7.43	47.45 ± 6.54	0.45 (−3.37 to 4.27)	0.989	−6.49 (−10.31 to −2.66)	<0.001	−6.93 (−10.75 to −3.11)	<0.001
Day 10	36.95 ± 6.82	36.71 ± 6.56	43.45 ± 6.61	0.34 (−3.30 to 3.98)	0.994	−6.34 (−9.98 to −2.71)	<0.001	−6.68 (−10.32 to −3.05)	<0.001
Additional dose released by PCA pump. H[C]: statistical magnitude									
Day 2	0.00 [0.00, 2.00]	2.00 [0.00, 2.00]	2.00 [2.00, 4.00]	H = −7.51	0.860	H = −27.62	<0.001	H = −20.11	0.013

Data are mean ± SD or median [Q1, Q3].
[a]Multiple comparison results adjusted using the Sidák correction.
[b]Time point at 6 h after surgery.
[c]Kruskal-Wallis one-way analysis of variance by ranks.
VAS: visual analog analogue scale; PCA: patient-controlled analgesia; CI: confidence interval; EA: electroacupuncture.

Figure 2. Visual analog scale (VAS) pain scores in the contralateral electroacupuncture (EA), ipsilateral EA and sham EA groups at each time point.



corresponding analgesic effects by regulating the release of neurotransmitters (such as endorphin and enkephalin) and blocking the pain signal pathway.[37,38] Liu et al.[39] explained that acupuncture at traditional acupuncture point locations could induce various somatosensory-autonomic nerve-target organ reflex pathways and regulate immunity/inflammation. However, the mechanism of action of contralateral needling has been rarely studied. In some types of pain, contralateral acupuncture may be more effective than ipsilateral acupuncture.[40] Other studies have confirmed that contralateral and ipsilateral acupuncture have similar analgesic effects,[41,42] but different underlying brain mechanisms.[20,22] The anterior cingulate cortex (ACC) is one of the candidate regions, and ACC neurons have a large bilateral receptive area of body sensory stimulation,[43] which means that contralateral acupuncture may competitively suppress noxious signals. However, ACC involvement in the emotional component of postoperative pain needs to be further investigated. Yang et al.[44] suggested that the analgesic effect mediated by the mu receptor during low-frequency contralateral EA pretreatment has an antinociceptive action against inflammatory pain, and it may provide a potential strategy to treat inflammatory arthritic pain.

The HSS scale includes the assessment of pain, joint function, and ROM, which account for 70% of the score.[45] From the comparison results of the three groups, joint function and ROM were improved with pain relief. After three interventions, the HSS scores of the two true EA groups were higher than the sham EA group but did not significantly differ between the two true EA groups. Postoperative pain may trigger a strong stress reaction, stimulate the release of inflammatory factors, and inhibit immune function.[46] EA at ST36 has been found to maintain immune function and improve postoperative recovery.[47] Studies have shown that peripheral opioids released by EA at the inflammatory site can alleviate the local inflammatory reaction, which is an effect that is unlikely to be seen with contralateral acupuncture.[48] Accordingly, in our trial, the improvement in the swelling around the knee joint after three interventions was significantly greater in the ipsilateral EA group than in the contralateral EA.

There were some limitations to our study. First, we did not conduct a long-term follow-up evaluation, and the follow-up period was restricted to the period of hospitalization. Therefore, further research is needed to explore the long-term effects of contralateral EA on pain and joint function rehabilitation after TKA. Second, referring to the trial of Liu et al.,[27] we attempted to blind the patients; a strict clinical controlled trial was designed and executed, which aimed to minimize the placebo effect. Sterile rubber pads were attached to the skin overlying the needling locations, and blunt needles were used through the adhesive pad without penetrating the skin; however, a slight pressure sensation was induced by contact with the blunt tip. Moreover, the wire inside the electrical stimulation apparatus was cut to ensure no current flowed through the connecting wires between the two needle handles, to create the appearance of real EA. We did not set a blinding method for acupuncturists. Ultimately, the expectation of participants and the communication between acupuncturists and participants may still have introduced performance bias. Third, the treatment cycle used in this study was short. Finally, as pain is a subjective experience, the effect of contralateral needling on postoperative pain should be objectively supplemented by further neuroimaging research.

Conclusion

In conclusion, both contralateral EA and ipsilateral EA methods improved pain and reduced opioid use after TKA,

Table 3. Characteristics and changes in secondary outcomes with multiple comparisons.

Outcome	Contralateral EA	Ipsilateral EA	Sham EA	Pairwise comparisons					
	Group A	Group B	Group C	Group A vs B		Group A vs C		Group B vs C	
				Effect size (95% CI)	P[a]	Effect size (95% CI)	P[a]	Effect size (95% CI)	P[a]
HSS score after surgery									
Day 3	61.43 ± 4.66	61.43 ± 4.61	57.95 ± 3.96	0.00 (−2.46 to 2.46)	1.000	3.48 (1.03 to 5.94)	0.003	3.48 (1.24 to 5.94)	0.003
Day 10	72.86 ± 4.20	74.29 ± 4.41	67.68 ± 3.47	−1.42 (−3.68 to 0.83)	0.337	5.18 (2.93 to 7.43)	<0.001	6.60 (4.35 to 8.85)	<0.001
AROM after surgery									
Day 3	57.00 ± 12.92	53.61 ± 13.17	52.89 ± 10.82	3.36 (−3.47 to 10.19)	0.554	3.93 (−2.91 to 10.77)	0.420	0.57 (−6.26 to 7.40)	0.996
Day 5	65.68 ± 12.19	64.89 ± 11.81	59.32 ± 11.35	0.73 (−5.68 to 7.14)	0.990	6.09 (−0.32 to 12.51)	0.068	5.36 (−1.05 to 11.77)	0.129
Day 7	74.71 ± 10.78	72.68 ± 12.25	66.74 ± 11.43	1.98 (−4.33 to 8.29)	0.832	7.75 (1.44 to 14.07)	0.011	5.77 (−0.54 to 12.09)	0.084
Day 10	89.18 ± 8.88	87.00 ± 11.57	80.53 ± 10.61	2.13 (−3.50 to 7.75)	0.739	8.40 (2.76 to 14.02)	0.001	6.26 (0.64 to 11.89)	0.024
PROM after surgery									
Day 3	67.47 ± 13.28	66.13 ± 13.23	59.82 ± 10.69	1.25 (−5.69 to 8.19)	0.962	7.47 (0.52 to 14.42)	0.031	6.22 (−0.72 to 13.16)	0.092
Day 5	76.21 ± 12.00	76.39 ± 13.12	69.34 ± 11.01	−0.38 (−7.01 to 6.25)	0.999	6.48 (−0.16 to 13.12)	0.058	6.86 (0.23 to 13.48)	0.040
Day 7	86.03 ± 10.13	84.58 ± 12.07	78.26 ± 11.28	1.30 (−4.88 to 7.48)	0.941	7.47 (1.28 to 13.66)	0.013	6.17 (−0.01 to 12.35)	0.051
Day 10	99.74 ± 7.24	99.82 ± 9.38	94.11 ± 8.87	−0.22 (−4.91 to 4.46)	0.999	5.35 (0.65 to 10.04)	0.020	5.57 (0.88 to 10.25)	0.014
Improvement of swelling around the knee joint after surgery[b]									
Day 3	0.011 ± 0.023	−0.002 ± 0.020	0.021 ± 0.018	0.014 (0.002 to 0.025)	0.012	−0.010 (−0.021 to 0.002)	0.145	−0.023 (−0.035 to −0.012)	<0.001
Day 5	0.010 ± 0.033	0.004 ± 0.036	0.024 ± 0.027	0.006 (−0.011 to 0.024)	1.000	−0.014 (−0.032 to 0.004)	0.167	−0.020 (−0.038 to −0.003)	0.019
Day 7	0.009 ± 0.030	0.014 ± 0.040	0.020 ± 0.026	−0.005 (−0.024 to 0.013)	1.000	−0.011 (−0.029 to 0.007)	0.430	−0.006 (−0.024 to 0.013)	1.000
Day 10	0.003 ± 0.025	0.005 ± 0.034	0.016 ± 0.024	−0.002 (−0.018 to 0.013)	1.000	−0.013 (−0.029 to 0.002)	0.119	−0.011 (−0.027 to 0.005)	0.258
HAMA score									
Day 3	12.53 ± 4.15	10.95 ± 2.73	13.76 ± 3.23	1.58 (−0.33 to 3.48)	0.134	−1.24 (−3.14 to 0.67)	0.314	−2.82 (−4.72 to −0.91)	0.002
Day 10	9.13 ± 3.41	7.87 ± 1.95	11.05 ± 2.11	1.26 (−0.17 to 2.69)	0.100	−1.92 (−3.35 to −0.49)	0.005	−3.18 (−4.62 to −1.75)	<0.001

Note: Data are mean ± SD.
[a]Multiple comparison results adjusted using the Sidák correction.
[b]According to the following formula: (COK on the Xth postoperative day —COK on the first postoperative day) / (COK on the first postoperative day).
HSS, Hospital for Special Surgery; PROM, passive range of motion; AROM, active range of motion; HAMA, Hamilton anxiety scale; CI, confidence interval; EA, electroacupuncture.

Table 4. Safety and other outcomes.

Variable		Contralateral EA (n=36)	Ipsilateral EA (n=37)	Sham EA (n=35)
Discomfort and acceptance of EA				
Discomfort of EA[a]	Day 1	19.97 ± 10.60	20.51 ± 10.12	20.94 ± 9.58
	Day 2	16.75 ± 9.26	16.78 ± 8.66	17.77 ± 8.95
	Day 3	12.25 ± 10.13	12.57 ± 8.91	16.11 ± 8.64
Acceptance of EA[a]	Day 1	84.44 ± 8.36	84.16 ± 9.39	83.06 ± 7.45
	Day 2	87.69 ± 6.75	87.57 ± 7.40	85.89 ± 8.37
	Day 3	90.97 ± 7.22	91.22 ± 7.32	87.26 ± 9.20
Assessment of blinding[b]		27 (71.05)	29 (76.32)	29 (76.32)
Side effects of acupuncture		2 (5.26)	1 (2.70)	0 (0.00)

Note: Data are mean ± SD or n (%).
[a]Refer to the visual analog score (VAS), measured using a range of 0–100.
[b]Cases lost to follow-up were considered to represent blinding failures.
EA: electroacupuncture.

in an effect that was already apparent after two rounds of the intervention. These findings increase our understanding of pain management after TKA, and the role of early intervention by nonpharmacologic treatment methods to promote postoperative rehabilitation.

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Contributors

Y.L.C. and L.B.X. conceived the study, designed the study protocol, and sought ethical approval. H.X. was the acupuncturist. X.L.S. and Y.Y.L. participated in the data evaluation and collection. L.Z. and K.M.T. were responsible for data entry and statistical analysis. H.H. drafted the article. K.M.T. and Y.L.C. participated in the critical revision of the article. All authors read and approved the final version of the article accepted for publication.

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Ethical approval and informed consent

All study participants provided written consent and received a full and detailed explanation of the trial process. The ethical validity of the study was assessed and approved by the Institutional Review Board of Guanghua Hospital, Shanghai University of Traditional Chinese Medicine (2018-K-18).

Consent for publication

Not applicable.

ORCID iDs

Hai Huang https://orcid.org/0000-0002-2376-0462
Xiuling Song https://orcid.org/0000-0002-3280-7189
Ling Zhao https://orcid.org/0000-0002-4017-8167

Supplemental material

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